A.A. 2024-25. Exercise 3: SEISMIC VULNERABILITY ASSESSMENT OF AN EXISTING MASONRY BUILDING

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ABSTRACT: This report presents a seismic assessment of the Caserta Vecchia Bell Tower through the simplified global analysis method (LV1), in accordance with the Guidelines for the Seismic Risk Assessment and Reduction of Cultural Heritage and NTC 2018 provisions. The tower is subdivided into eleven parts with assumed constant cross-section to evaluate the seismic response under horizontal actions. For each section, the acting and resisting bending moments were calculated in both principal directions (X and Y), allowing for the estimation of the collapse multiplier, the ground acceleration capacity, and the corresponding seismic safety index. In addition to the global analysis, a local LV2 evaluation was conducted on the most critical portion of the structure — the base section along the Y-axis — where the presence of a large arch significantly reduces the resisting capacity. A linear kinematic mechanism was modeled, and the load multiplier triggering local failure was computed. The comparison between acting and resisting moments confirmed the need for strengthening measures at the base. Overall, the tower shows satisfactory global seismic performance but exhibits local vulnerabilities that may compromise its safety without targeted intervention.

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1. INTRODUCTION

This report presents the seismic assessment of the Caserta Vecchia Bell Tower, applying a Level 1 simplified global analysis (LV1) in accordance with the Guidelines for the evaluation and mitigation of seismic risk for cultural heritage (D.M. 14/01/2008 and related Circular). These provisions allow for interventions focused on improving seismic performance, provided they are supported by appropriate safety assessments.

The Caserta Vecchia bell tower, due to its geometric regularity and symmetry, offers favorable conditions for this type of analysis, as the center of mass and center of stiffness are assumed to coincide with the geometric center of the base section. This condition simplifies the global behavior of the structure under seismic action, allowing a more direct evaluation of its capacity.

The primary objective of this study is to verify the seismic resistance of the structure by comparing the acting and resistant moments in both principal directions (x and y), following the press-flexure verification method defined by the guidelines. The tower is subdivided into distinct vertical segments, and for each section the seismic capacity is determined by evaluating the moment of resistance based on the structural geometry and material properties.

This process ultimately leads to the definition of a safety factor, based on the comparison between the ground acceleration capacity and the expected seismic demand at the site. The results help identify the most vulnerable sections of the tower and support the decision-making process for possible mitigation measures.

Figure 1 – View of The Saint Michele Arcangelo Cathedral in Caserta Vecchia

1.1 Historical and architectural background

The bell tower of Caserta Vecchia is part of the Romanesque complex of the Cathedral of San Michele Arcangelo, located in the medieval village of Caserta Vecchia, in southern Italy. The tower dates back to the 12th century and is one of the most significant examples of Romanesque architecture in Campania, characterized by a blend of Norman, Arabic, and Byzantine influences.

The tower stands approximately 31.5 meters tall, with a square base transitioning to an octagonal upper section, a typology often found in Romanesque bell towers of southern Italy. Its base level includes a large pointed arch, which historically served as a passageway under the tower, suggesting both symbolic and functional roles in the urban layout of the village.

Over the centuries, the tower has undergone limited structural transformations, maintaining much of its original configuration. It is constructed primarily of local tuff and limestone, with visible signs of material degradation and localized cracking due to aging, environmental exposure, and possible seismic events. Despite its preservation, the structure shows vulnerabilities typical of slender masonry towers, such as high compressive stress concentrations and potential for flexural collapse mechanisms under seismic loading.

This historical context underscores the importance of assessing the tower's seismic performance within the broader goal of preserving Italy's cultural heritage. The current study aims to contribute to this effort by providing a structural analysis that is consistent with the methodologies recommended for heritage buildings.

(a) (b)

Figure 2 - (a) Drone photogrammetric survey; (b) Terrestrial photogrammetry and point cloud in Agisoft Photoscan

2. STRUCTURAL AND GEOMETRIC CHARACTERIZATION

2.1. Overview of the Caserta Vecchia Bell Tower

The bell tower of the San Michele Arcangelo Cathedral in Caserta Vecchia, Italy, is a significant 13th-century architectural and historical landmark. Located in Campania, this structure is square-based with a base side length of 7.50 meters and a total height of 31.5 meters. It is notably divided into five distinct levels, each delineated by pronounced cornices, which contribute to a gradual transition in visual weight and structural emphasis from the base upwards.

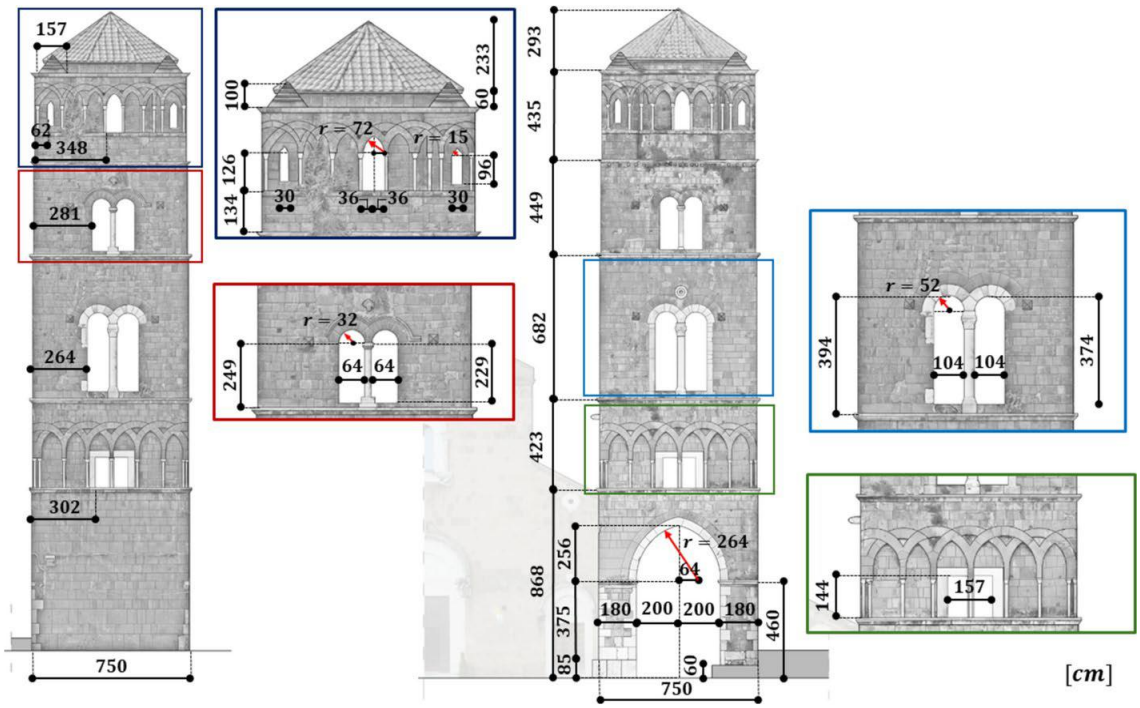

Figure 3 - Facades of the tower.

A particularly noteworthy feature of the tower is its transition from a square to an octagonal geometry at its pinnacle, achieved through the addition of cylindrical turrets at the corners. The tower's base is visible due to the lowered adjacent square level and incorporates an acute arched opening, seamlessly blending architectural elegance with urban functionality. Each level exhibits specific design elements: the second level features limestone-framed rectangular openings, while the third and fourth levels are adorned with biforas. The façade of the tower reflects Arab-Norman art influences, characterized by acute arched openings and biforas framed by relief arches and pilasters.

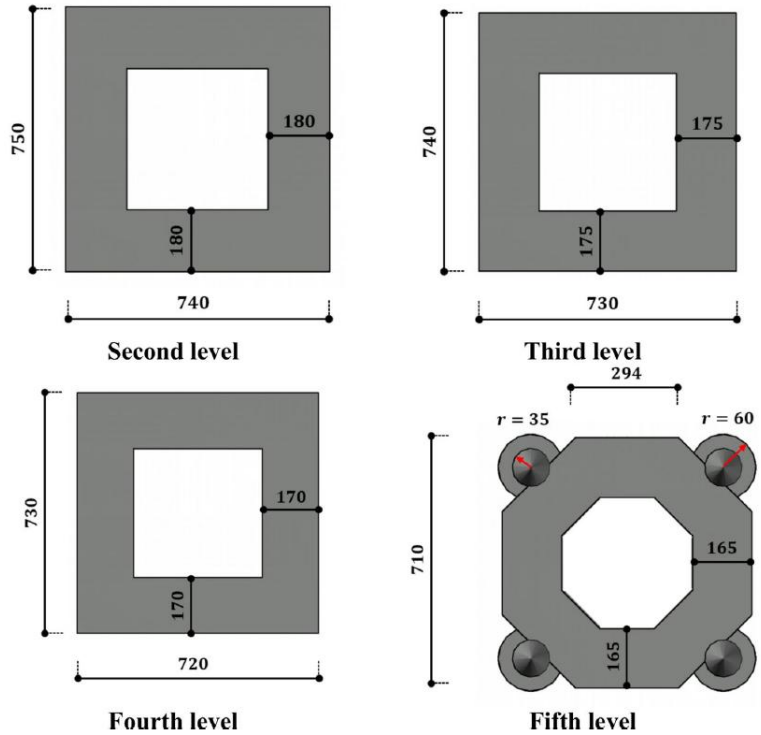

Figure 4 - Detailed floor plan dimension [cm]

2.2. Geometric assumptions and modeling strategy

For the evaluation of the seismic risk of the buildings, three analysis levels can be performed. Each of them depends on the level of analysis required, and the scale in which they need to applied. These levels of analysis are as follows:
-LV1 – qualitative analysis and evaluation with simplified mechanical models. Approximate global analysis of the artefact for the evaluation of seismic safety to be used in territorial scale.
-LV2 – analysis for the assessments in the presence of local interventions on limited areas of the artefact.
-LV3 – analysis for the assessment of interventions that affect the artefact as an entirety or when a particular assessment of the seismic safety is required.

In our case, the seismic vulnerability assessment of the building shall be made by the analysis level LV1, as proposed by the DPCM 09/02/11 in the par. 5.4.4., "Torri, campanili ed altre strutture a prevalente sviluppo verticale". The seismic assessment of the tower will employ the Level LV1 simplified global analysis method, as prescribed by the guidelines for cultural heritage. This approach is designed for territorial-scale evaluations of seismic safety and relies on simplified models. LV1 models are typically based on a limited number of geometric and mechanical parameters, or they can utilize qualitative data obtained from visual inspections and constructive characteristics. For structures like bell towers, which primarily exhibit a vertical development, the LV1 model often conceptualizes the tower as a cantilever, focusing on collapse by flexural-compression.

This method provides a preliminary qualitative judgment on the level of risk, specifically seismic risk.

It involves a rigorous yet agile collection of information, often through standardized forms, to ascertain the structural behavior and identify potential collapse mechanisms. The application of simplified models, even with inherent uncertainties, offers the advantage of providing a homogeneous and comparable evaluation across a territory, which is crucial for planning interventions.

2.3. Soil condition and topography

The seismic analysis requires defining the local soil conditions to properly characterize the site's response to ground shaking. Based on a visual survey of the surrounding geological context, and in the absence of site-specific geotechnical investigations or stratigraphic profiles, the subsoil at the base of the Caserta Vecchia bell tower has been classified as Category C, according to Table 3.2.II of the NTC 2018.

Category C corresponds to a layer of medium-dense or medium-stiff materials, such as sands, silts, clays, or gravels, with a thickness greater than 30 meters overlying more competent rock. The choice of this classification is justified by the following observations: the tower is situated on a natural hilltop, but without sharp morphological or lithological transitions typically associated with rocky outcrops (Category A) or deep soft layers (Category D/E); the historical urban fabric of Caserta Vecchia was developed on sedimentary substrates typical of inland Campania, often composed of weathered tuff, marly layers, or compact clays, which align with the expected profile for Category C; a conservative approach excludes more favorable categories (A or B) in the absence of shear wave velocity data, while Categories D and E would not represent the site's stiffness conditions unless soft, compressible layers were explicitly detected.
This assumption ensures a realistic yet cautious representation of the soil stiffness, in line with standard practice when dealing with historic structures and limited subsoil information.

The classification of the subsoil as Category C directly influences the spectral shape factors used to define the elastic response spectrum for seismic analysis. In particular, this classification determines the values of the coefficients Ss and Cc, which govern the horizontal elastic response spectrum.

Categoria sottosuolo	S_S	C_C
A	1,00	1,00
B	$1{,}00 \le 1{,}40 - 0{,}40 \cdot F_o \cdot \frac{a_g}{g} \le 1{,}20$	$1{,}10 \cdot (T_C^*)^{-0{,}20}$
C	$1{,}00 \le 1{,}70 - 0{,}60 \cdot F_o \cdot \frac{a_g}{g} \le 1{,}50$	$1{,}05 \cdot (T_C^*)^{-0{,}33}$
D	$0{,}90 \le 2{,}40 - 1{,}50 \cdot F_o \cdot \frac{a_g}{g} \le 1{,}80$	$1{,}25 \cdot (T_C^*)^{-0{,}50}$
E	$1{,}00 \le 2{,}00 - 1{,}10 \cdot F_o \cdot \frac{a_g}{g} \le 1{,}60$	$1{,}15 \cdot (T_C^*)^{-0{,}40}$

Figure 5 - Table 3.2.IV from NTC18

The Caserta Vecchia bell tower is located on a moderately elevated site, yet the surrounding terrain does not exhibit steep slopes or significant morphological discontinuities. In accordance with Table 3.2.III of NTC 2018, the area can be classified as Topographic Category T1, which corresponds to flat or mildly sloping ground with an average inclination of less than 15°.

Topographic effects are accounted for in seismic analysis through the topographic amplification factor ST, which varies depending on the site's configuration. For Category T1, the amplification factor is ST equal 1, meaning that no additional amplification due to topographic conditions is applied to the seismic input.

2.3. Subdivision of the tower

According to the geometry of the tower, it was divided into segments with identical cross-sections. In total, 11 sections were considered: two for each level, and an additional one for the roof. The full (solid) section has been previously shown in Figure 4; below, we present the perforated sections, which result from the presence of biforas.

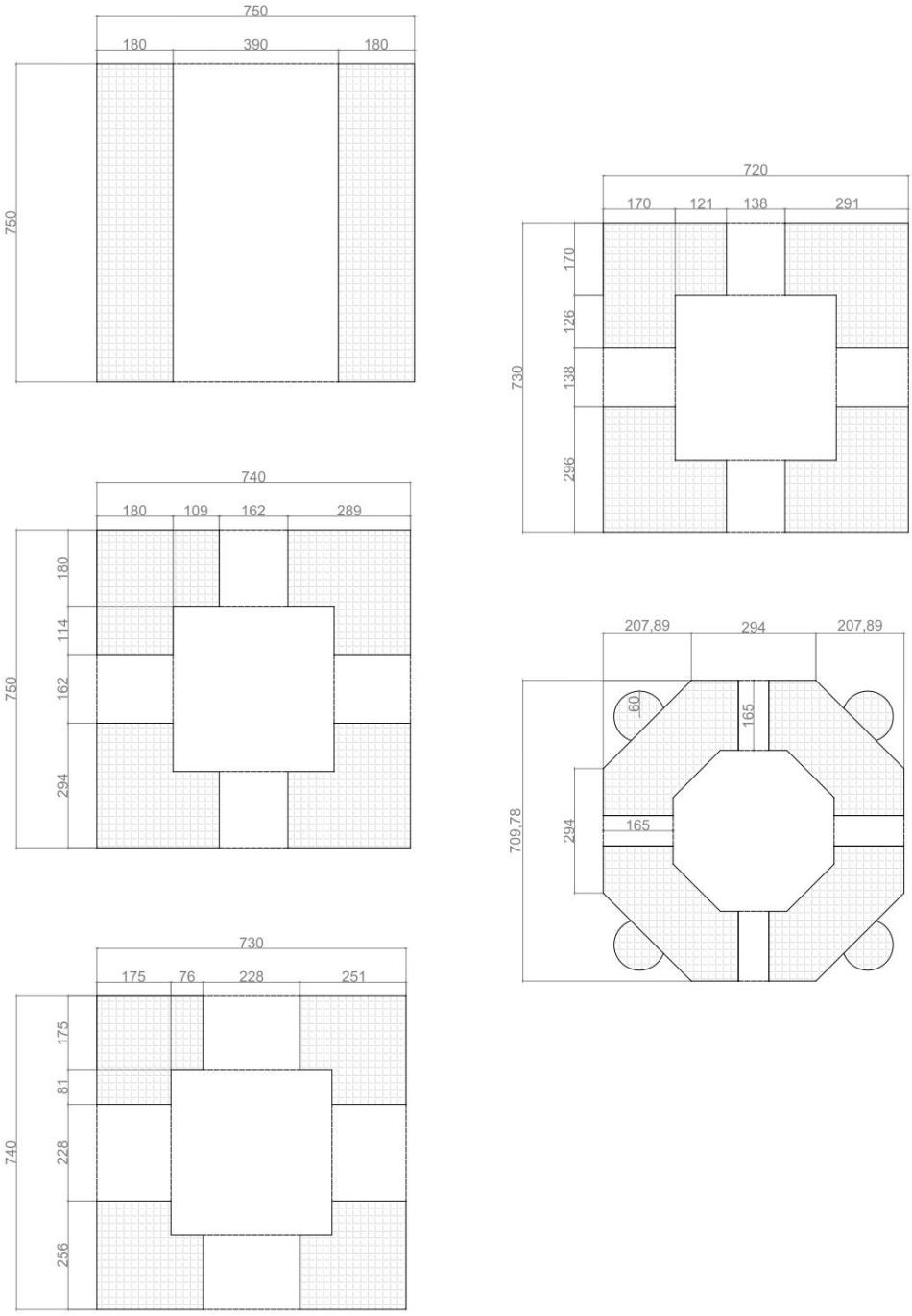

Figure 6 - Left column from top to bottom: perforated sections of the base, first, and second floors. Right column from top to bottom: perforated sections of the third and fourth floors.

2.4. Mechanical Properties and Material Data

The masonry of the Caserta Vecchia bell tower, based on visual inspection and construction typology, is best classified as irregular masonry of soft stone (muratura irregolare di pietra tenera), rather than disarranged rubble masonry. This choice is supported by the relatively regular stratification observed in the external faces, the typical use of soft stones such as tuff or calcarenite in Campania, and the absence of large, round, uncut cobbles commonly associated with disarranged rubble.

According to the Italian technical standards (NTC 2018 and Circolare 7/2019, Table C8.5.I), this typology is characterized by better average mechanical performance and lower density, which is favorable for seismic behavior.

Given the limited documentation and lack of in-situ testing, the Knowledge Level (LC) is classified as LC1, and a Confidence Factor is adopted.

The mechanical properties used for the analysis are defined in the following table.

Table 1 - Adopted mechanical properties

Property	Value	Unit	Notes
Characteristic compressive strength (f)	1.4	N/mm²	The design value is diffrent
Confidence factor (Fc)	1.35	–	LC1 level
Partial safety factor (γ_m)	3.0	–	Circolare 7/2019
Elastic modulus (E)	1080	N/mm²	Mean value
Shear modulus (G)	360	N/mm²	Mean value
Unit weight (w)	16	kN/m³	Standard value for tuff/rubble masonry (light rocks)

3. LOAD ANALYSIS

3.1. Permanent structural loads

The self-weight loads were calculated by determining the volume of masonry for each vertical section (i), and then multiplying by the unit weight (16 kN/m³) to obtain the loads in kilonewtons (kN).

An additional load was included at each floor level to account for the presence of a timber floor slab, assumed to have a thickness of 0.15 m and a unit weight of 5 kN/m³.

Further contributions to the total weight include: the arched structure connected to the first level, the small columns (colonette) of the bifora at the third level and finally the roof which has been recently rebuilt in reinforced concrete.

Table 2 – Values of self-weight

	i	hi [m]	Ai [m²]	Vi [m³]	Wi ' [kN]	W added [kN]	Wi [kN]
BS	1	7.16	27.00	193.32	3093.1	19.80	3112.9
	2	1.52	41.04	62.38	998.1	0.00	998.1
1S	3	1.50	29.02	43.52	696.4	40.52	736.9
	4	2.73	40.68	111.06	1776.9	0.00	1776.9
2S	5	3.74	23.24	86.92	1390.7	41.77	1432.5
	6	3.08	39.20	120.74	1931.8	0.00	1931.8
3S	7	2.29	28.36	64.94	1039.0	37.80	1076.8
	8	2.20	37.74	83.03	1328.4	0.00	1328.4
4S	9	1.34	29.83	39.98	639.6	31.30	670.9
	10	1.26	23.10	29.11	465.7	0.00	465.7
	11	1.75	29.83	52.21	835.3	5.71	841.0

The total weight of the building is 14372.0 kN.

3.1. Seismic Input and Return Periods

The seismic action for the tower is defined using the reference elastic response spectrum, as prescribed by Annex 1 of the NTC 2018. This spectrum is constructed based on the geographic coordinates of the site and the corresponding seismic hazard values provided by the national database, without the need for site-specific probabilistic seismic hazard analysis.

The material used for the tower's structure is stone masonry, with some presence of brick, and is modeled with a unit weight of 16 kN/m³, consistent with historical masonry typologies in Campania.

In the case of heritage structures, the guidelines allow the reduction of structural interventions, provided that a safety assessment is performed in accordance with the relevant limit states. This requires defining the nominal life VN and the use class CU, which together determine the reference period VR for seismic action.

For this project the Nominal life VN is 50 years and the use class, CU, is 1 (corresponding to Class II – frequent use with normal occupancy).

	TIPI DI COSTRUZIONE	Vita Nominale V_N (in anni)
1	Opere provvisorie – Opere provvisionali - Strutture in fase costruttiva[1]	≤ 10
2	Opere ordinarie, ponti, opere infrastrutturali e dighe di dimensioni contenute o di importanza normale	≥ 50
3	Grandi opere, ponti, opere infrastrutturali e dighe di grandi dimensioni o di importanza strategica	≥ 100

Figure 7 - Nominal life

Classe I: Costruzioni con presenza solo occasionale di persone, edifici agricoli.

Classe II: Costruzioni il cui uso preveda normali affollamenti, senza contenuti pericolosi per l'ambiente e senza funzioni pubbliche e sociali essenziali. Industrie con attività non pericolose per l'ambiente. Ponti, opere infrastrutturali, reti viarie non ricadenti in Classe d'uso *III* o in Classe d'uso *IV*, reti ferroviarie la cui interruzione non provochi situazioni di emergenza. Dighe il cui collasso non provochi conseguenze rilevanti.

Classe III: Costruzioni il cui uso preveda affollamenti significativi. Industrie con attività pericolose per l'ambiente. Reti viarie extraurbane non ricadenti in Classe d'uso *IV*. Ponti e reti ferroviarie la cui interruzione provochi situazioni di emergenza. Dighe rilevanti per le conseguenze di un loro eventuale collasso.

Classe IV: Costruzioni con funzioni pubbliche o strategiche importanti, anche con riferimento alla gestione della protezione civile in caso di calamità. Industrie con attività particolarmente pericolose per l'ambiente. Reti viarie di tipo A o B, di cui al D.M. 5 novembre 2001, n. 6792, "Norme funzionali e geometriche per la costruzione delle strade", e di tipo C quando appartenenti ad itinerari di collegamento tra capoluoghi di provincia non altresì serviti da strade di tipo A o B. Ponti e reti ferroviarie di importanza critica per il mantenimento delle vie di comunicazione, particolarmente dopo un evento sismico. Dighe connesse al funzionamento di acquedotti e a impianti di produzione di energia elettrica.

Figure 8 - Construction Classes.

Reference period:

$$V_R = V_N \cdot C_U = 50 \cdot 1 = 50\ years$$

The return periods TR for each limit state are calculated based on the following formula:

$$T_R = -V_R / \ln(1 - P_{VR})$$

where PVR is the probability of exceedance associated with each performance level.

Table 3 - Values used for the seismic assessment, according to the Limit State.

	PVR [%]	TR [years]	ag [g]	F0 [-]	Tc* [s]
SLO	81	30.107	0.043	2.377	0.285
SLD	63	50.289	0.054	2.368	0.320
SLV	10	474.561	0.121	2.517	0.419
SLC	5	974.786	0.150	2.575	0.437

Where ag is the maximum horizontal ground acceleration, F0 is the peak amplification factor of the horizontal acceleration spectrum, and Tc∗ is the start period of the constant-velocity segment of the spectrum. These values define the site-specific seismic input, used to construct the elastic response spectra and to evaluate the tower's seismic capacity in relation to the expected demand at each limit state. The selected parameters are consistent with those provided in Table 2 of Annex B of the NTC 2018.

			T_R=30			T_R=50			T_R=72			T_R=101			T_R=140			T_R=201			T_R=475			T_R=975			T_R=2475		
ID	LON	LAT	a_g	F_o	T_C^*	a_g	F_o	T_C^*	a_g	F_o	T_C^*	a_g	F_o	T_C^*	a_g	F_o	T_C^*	a_g	F_o	T_C^*	a_g	F_o	T_C^*	a_g	F_o	T_C^*	a_g	F_o	T_C^*
32090	14.22	41.083	0.423	2.39	0.28	0.530	2.37	0.32	0.606	2.43	0.34	0.694	2.44	0.35	0.779	2.47	0.36	0.885	2.50	0.38	1.159	2.55	0.43	1.414	2.63	0.45	1.779	2.74	0.51
31868	14.22	41.133	0.424	2.43	0.29	0.525	2.43	0.32	0.596	2.50	0.34	0.682	2.51	0.35	0.766	2.54	0.37	0.871	2.52	0.40	1.155	2.59	0.44	1.433	2.65	0.46	1.840	2.74	0.52
31646	14.221	41.183	0.433	2.45	0.29	0.532	2.48	0.32	0.603	2.55	0.34	0.692	2.55	0.36	0.777	2.59	0.37	0.891	2.55	0.41	1.208	2.59	0.44	1.537	2.62	0.46	2.026	2.69	0.52
31424	14.222	41.233	0.450	2.46	0.29	0.551	2.51	0.32	0.631	2.56	0.34	0.724	2.57	0.35	0.819	2.60	0.37	0.949	2.54	0.40	1.325	2.55	0.44	1.715	2.57	0.45	2.325	2.64	0.47

Figure 9 – Seismic parameters from NTC corresponding to the reference point closest to the site (Caserta).

3.3. Horizontal elastic response spectrum

The horizontal elastic response spectrum was constructed in accordance with the provisions of NTC 2018, specifically following the 3.2.3.2.1 paragraph of NTC. The spectrum defines the seismic demand in terms of spectral acceleration Se(T) as a function of the vibration period T, and accounts for local site conditions through parameters such as the design ground acceleration ag, the amplification factor F0, the corner period Tc∗, and the slope reduction factor η.

$$S_e(T) = \begin{cases} a_g \cdot S \cdot \eta \cdot F_o \cdot \left[\frac{T}{T_B} + \frac{1}{\eta \cdot F_o}\left(1 - \frac{T}{T_B}\right)\right] & for \quad 0 \le T < T_B \\ a_g \cdot S \cdot \eta \cdot F_o \quad for \quad T_B \le T < T_C \\ a_g \cdot S \cdot \eta \cdot F_o \cdot \left(\frac{T_C}{T}\right) \quad for \quad T_C \le T < T_D \\ a_g \cdot S \cdot \eta \cdot F_o \cdot \left(\frac{T_C \cdot T_D}{T^2}\right) \quad for \quad T_D \le T \end{cases}$$

Figure 10 - Horizontal elastic response spectrum system.

Se(T) is the elastic acceleration response spectrum of the horizontal component; T is the period of vibration; S is the coefficient accounting for subsoil category and topographical conditions; η is the damping correction factor for the elastic spectrum that is equal to 1, as a damping equal to 5% has been assumed; F0 is the maximum spectral amplification factor for horizontal acceleration; ag is the design ground acceleration at the site; Tc∗ is the reference period marking the start of the constant velocity range.

In accordance with NTC 2018, the values of the spectral coefficients S_S and S_T, which contribute to the calculation of the site amplification factor S, depend on the subsoil category and the limit state under consideration.

Since the subsoil has been classified as Category C, the value of S_S reaches its maximum of 1.5 for the first three limit states (SLO, SLD, SLV); on the other hand, the value of S_T is equal to 1 cause of T1 topography. Also in the time intervals, the C soil category defines the Cc coefficient that varies slightly with each limit state, as it is a function of the period Tc ∗, which increases with more severe performance levels.

Table 4 – Values of Elastic Response Spectrum at Limit States.

	Tr [years]	ag [g]	F0 [-]	η [-]	S [1/g]	Tb [s]	Tc* [s]	Tc [s]	Td [s]
SLO	30.107	0.043	2.377	1.000	1.500	0.151	0.285	0.453	1.772
SLD	50.289	0.054	2.369	1.000	1.500	0.147	0.320	0.442	1.816
SLV	474.561	0.121	2.517	1.000	1.500	0.193	0.419	0.578	2.084
SLC	974.786	0.150	2.575	1.000	1.468	0.201	0.437	0.603	2.200

These values are used to build the spectral curve specific to each limit state, reflecting the growing demand as the return period increases.

This approach ensures that the seismic demand is properly scaled for different levels of structural performance, while maintaining consistency with the national code and the selected geotechnical classification.

3.4. Behaviour factor and design response spectrum

According to the Guidelines for the Seismic Risk Assessment and Reduction of Cultural Heritage and with reference to NTC 2018, Section 5.4.2, a behaviour factor q=2.95 has been adopted for

the present analysis. This value reflects the structural characteristics of the bell tower and is consistent with the recommendations provided for ordinary masonry structures.

In particular, as specified in NTC 2018, §7.8.1.3, the behaviour factor q for masonry buildings can be determined based on the formula:

$$q = 1.75(\alpha_U/\alpha_1)$$

Where α_U/α_1 is the ratio between the ultimate and first yield base shear. For regular, vertically bearing masonry structures without excessive irregularities, this ratio is typically assumed to be 1.7. The result of q is therefore justified by the tower's vertical development, regular geometry, and the absence of significant structural discontinuities or torsional irregularities.

Finally, the horizontal response spectrum can be observed in the figure below.

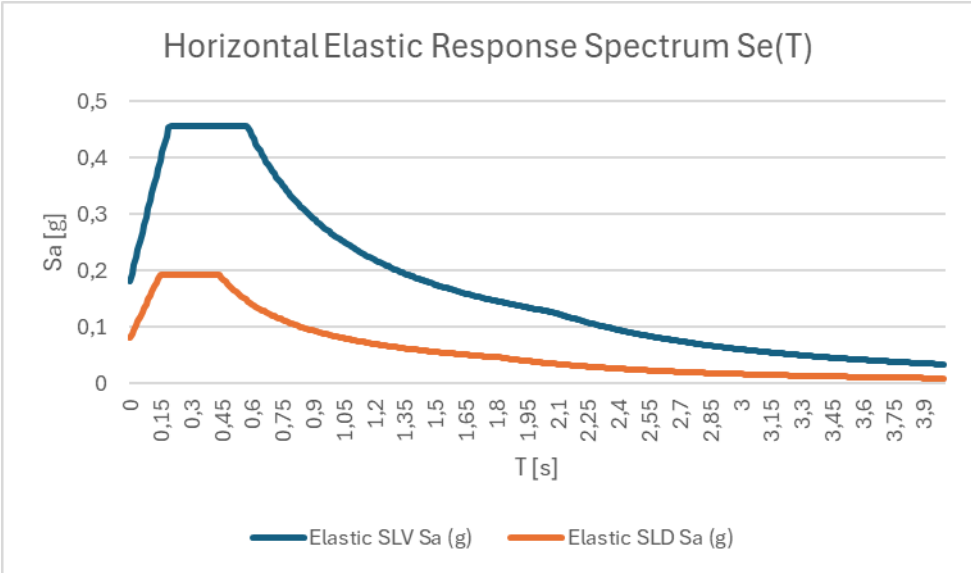

Figure 11 - Horizontal response spectrum

4. SEISMIC RISK ASSESMENT

4.1. Fundamental period and seismic base shear

The fundamental period of a structure is a key parameter in seismic analysis, as it characterizes the dynamic response of the building under earthquake loading. For the simplified global analysis (LV1) of a bell tower, the structure can be idealized as a vertical cantilever beam with a concentrated mass at the top that corresponds to 1/3 of the total mass.

Under this assumption, the fundamental period T is calculated using the following expression:

$$T = \frac{2\pi}{3}\sqrt{\frac{MH^3}{EI}}$$

Where: T, fundamental period of the structure; M, total mass of the tower; E, Young's modulus of masonry; I, moment of inertia of the base section (calculated via CAD); H: total height of the structure.

For the Caserta Vecchia bell tower, the height is 28.57 m, and the total self-weight has been calculated as 14372.0 kN, resulting in a total mass M of 1465.03 tons. The moments of inertia in the two principal directions were determined from the geometric model using AutoCAD:

$$Ix = 181.125\, m^4 \quad Iy = 203.331\, m^4$$

The corresponding spectral pseudo-accelerations Se(T) were obtained from the SLV elastic spectrum, derived according to NTC2018 using the appropriate parameters for the site in Caserta.

Once the spectral acceleration is known, the seismic base shear Fh can be computed using the simplified formula:

$$Fh = 0.85 \cdot Se(T) \cdot W/q$$

Where W is total weight of the tower and q is the behavour factor. This approach provides a quantitative estimate of the seismic demand acting at the base of the tower.

Table 5 – Fundamental periods and seismic base shear (Fh) of the tower.

	T [s]	Se (T) [g]	Fh [kN]
Dir X	0.826	0.316	1296.6
Dir Y	0.875	0.298	1224.0

The results show that, due to the high total weight and moderately long vibration period, the Caserta Vecchia bell tower is subjected to significant horizontal forces, highlighting the importance of assessing its capacity to resist flexural-compression mechanisms and the possible need for strengthening interventions.

5. ACTING AND RESISTING MOMENT

In the seismic assessment of masonry towers, the concepts of acting moment (also referred to as overturning moment) and resisting moment (or stabilizing moment) are essential to evaluate the stability of the structure and to identify potential collapse mechanisms. The acting moment represents the overturning or flexural effect caused by horizontal seismic forces combined with the vertical loads applied to the structure. These forces tend to induce rotation or failure of the system. In contrast, the resisting moment reflects the tower's capacity to counteract this overturning, mainly through its self-weight, vertical loads, and the mechanical resistance of the masonry.

According to Section 5.4.4 of the Guidelines for the Seismic Risk Assessment and Reduction of Cultural Heritage (NTC 2018), this verification must be conducted in both main directions of the structure (typically X and Y).

5.1. Acting moment

To compute the acting moment within a simplified mechanical model, the tower is idealized as a cantilever beam subjected to its own self-weight and to distributed horizontal forces. The structure is subdivided into multiple segments with uniform geometric properties to facilitate the analysis.

The total horizontal force Fh, acting on the base of the structure, has been calculated previously and it's distributed among the segments based on their individual weight Wi and the height zi of their center of mass above the foundation.

$$F_i = \frac{W_i \cdot z_i}{\sum_{k=1}^{n} W_k \cdot z_k} \cdot F_h$$

Where: W_i the weight of segment i; z_i the vertical coordinate of the centre of mass of a segment; F_h the total base shear.

The overall acting moment is calculated for each vertical section i of the tower, using the following expression:

$$M_i = \frac{\sum_{k=i}^{n} W_k z_k^2 - z_i^* \sum_{k=i}^{n} W_k z_k}{\sum_{i=1}^{n} W_k z_k} \cdot F_h$$

Where: zi∗ is the height at the base of section i, i.e., the elevation from which the section starts; zk is the vertical position of the center of mass, where the horizontal force Fk (previously find as Fi) is assumed to act; Wk is the vertical load (self-weight) acting on section I; Fh is the total seismic horizontal force acting in the considered direction (either X or Y).

This formulation accounts for the contribution of all segments above and including section i, considering both their weight and lever arm relative to the base of the section.

To compute the bending moment in direction X, the previously calculated horizontal force Fh,x must be used in the equation, yielding the acting moment about the Y axis (i.e., My). Conversely,

to compute the bending moment in direction Y, the horizontal force Fh,y is used, resulting in the acting moment about the X axis (i.e., Mx).

This step is fundamental in the simplified LV1 approach, as it allows identifying the most critical section—i.e., the one where the acting moment reaches or exceeds the resisting capacity—thus governing the verification of seismic safety.

Table 6 - Data and result of acting bending moment.

i	z_i^* [m]	z_k [m]	W_k [kN]	My [kN m]	Mx [kN m]
1	0.00	3.58	3112.9	24389.373	23025.0
2	7.16	7.92	998.1	15354.419	14495.4
3	8.68	9.43	736.9	13526.541	12769.8
4	10.18	11.55	1776.9	11792.088	11132.4
5	12.91	14.78	1432.5	8868.8337	8372.7
6	16.65	18.19	1931.8	5349.5245	5050.3
7	19.73	20.88	1076.8	2991.2991	2824.0
8	22.02	23.12	1328.4	1648.7692	1556.5
9	24.22	24.89	670.9	723.36863	682.9
10	25.56	26.19	465.7	357.53978	337.5
11	26.82	27.70	841.0	126.92074	119.8

5.2. Resisting moment

To evaluate the ultimate resisting bending moment, it is essential to consider the nonlinear mechanical behavior of masonry, which is characterized by zero tensile strength and effective resistance only in compression.

In a symmetric section, when the structure is subjected to flexural-compression, the eccentricity of the load leads to a partial activation of the cross-section: only the compressed region contributes to the resisting moment, while the remaining portion is ineffective due to the material's inability to resist tension.

The first step in the calculation is to determine the compressed length x of the section, obtained by imposing vertical equilibrium between the axial load and the compressive stress distribution. This length is computed using the formula:

$$x = \frac{W}{0.85 \cdot f_d \cdot a}$$

Where: x is the length of the compressed portion of the section in the analyzed direction; W is the axial load acting on the section (total vertical self-weight above); f_d is the design compressive strength of the masonry; a is the thickness of the section orthogonal to the analyzed direction (if x is along the X-axis, a is the thickness along Y).

Once x is known, the ultimate resisting moment is calculated as:

$$\mathrm{M} = \frac{\mathrm{W}}{2} \cdot (b - x)$$

Where: Mres is the resisting bending moment in the considered direction; W is the vertical load applied to the section; b is the total width of the section in the analyzed direction (including possible voids); x is the compressed length, as defined above; (b−x) represents the lever arm between the vertical force and the compressed resultant.

This process must be repeated for both principal directions (X and Y) to evaluate the resisting capacity of the tower under seismic actions acting in either direction. Geometric voids such as biforas must be included in the section's definition to accurately reflect the available resisting area.

Table 7 - Ultimate resisting bending moments in the Y direction.

i	Wi [kN]	a [m]	x [m]	b [m]	ex [m]	Mry [kN m]
1	14372.0	7.50	4.73	7.50	2.77	39844.7
2	11259.0	7.50	3.70	7.50	5.65	31796.6
3	10260.9	5.88	4.31	7.50	5.35	27434.6
4	9524.0	7.40	3.18	7.50	5.91	28154.9
5	7747.1	5.02	3.81	7.40	5.50	21290.4
6	6314.7	7.30	2.13	7.40	6.33	19995.3
7	4382.9	5.82	1.86	7.30	6.37	13961.8
8	3306.1	7.20	1.13	7.30	6.73	11131.1
9	1977.7	5.02	0.97	7.10	6.61	6538.0
10	1306.8	5.02	0.64	7.10	6.78	4427.7
11	841.0	5.02	0.41	7.10	6.89	2897.8

Table 8 - Ultimate resisting bending moments in the X direction.

i	Wi [kN]	a [m]	x [m]	b [m]	ex [m]	Mrx [kN m]
1	14372.0	3.60	9.85	7.50	-	-
2	11259.0	7.50	3.70	7.50	5.65	31796.6
3	10260.9	5.88	4.31	7.40	5.25	26921.5
4	9524.0	7.50	3.13	7.40	5.83	27779.5
5	7747.1	5.02	3.81	7.30	5.40	20903.0
6	6314.7	7.40	2.11	7.30	6.25	19725.1
7	4382.9	5.82	1.86	7.20	6.27	13742.7
8	3306.1	7.30	1.12	7.20	6.64	10978.6
9	1977.7	5.02	0.97	7.10	6.61	6538.0
10	1306.8	5.02	0.64	7.10	6.78	4427.7
11	841.0	5.02	0.41	7.10	6.89	2897.8

5.3. Analysis of the results

During the calculation of the ultimate resisting bending moment for each section of the tower, some critical observations emerged regarding the geometric consistency and the physical admissibility of the results. In particular, at the base of the tower, where vertical loads are highest, the compressed length x, derived from the vertical equilibrium, was found to be greater than the total width b of the section along the y axis. This situation occurs when the orthogonal thickness a is very small and leads to a non-physical result that violates the no-tension behavior assumed for masonry: a compressed length larger than the section itself implies an unrealistic stress distribution.

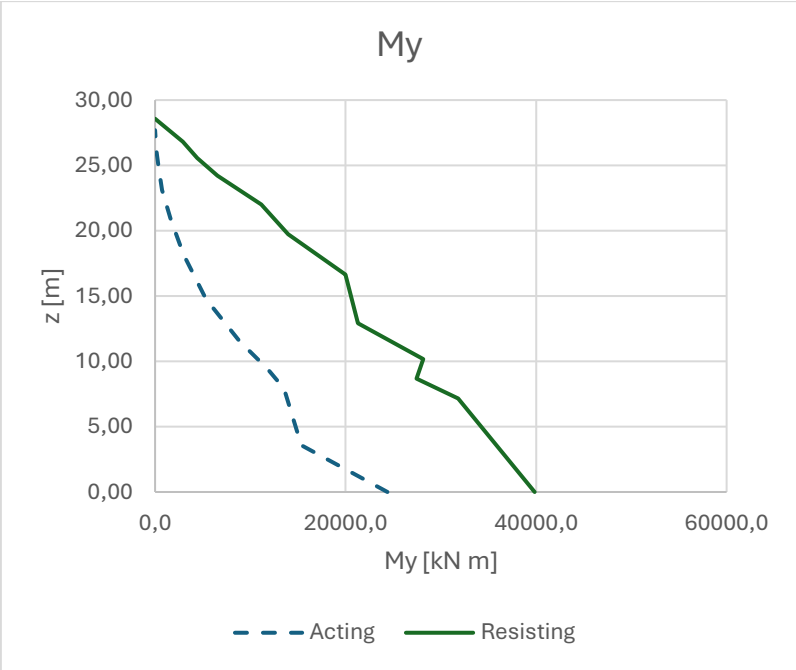

Figure 12 - Acting and resisting bending moments (My) along the height of the tower.

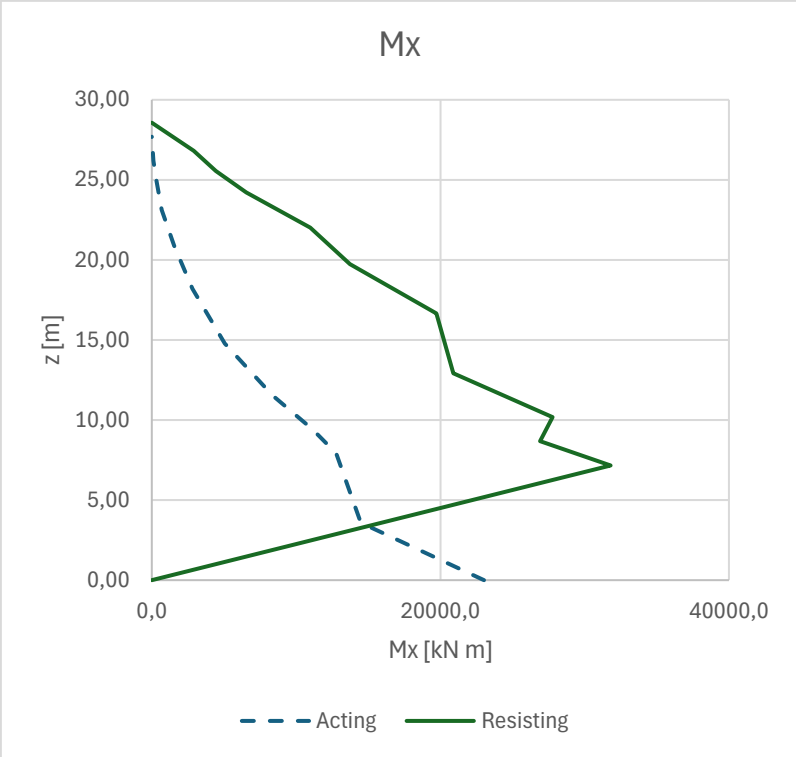

Figure 13 - Acting and resisting bending moments (Mx) along the height of the tower.

In the graphical representation and tabulated results, the resisting moment, Mrx, at the base section was initially shown as zero, since the calculated compressed length exceeded the section width under the assumption of reduced wall thickness. However, it must be noted that this condition is not realistic in the presence of a square foundation slab, where the dimensions along both axes can be considered equal (i.e., a=b). This geometric symmetry ensures that the compressed length remains within the limits of the section, and the resulting resisting moment is

significantly higher than the acting moment. Therefore, the apparent failure indicated in that section does not reflect the actual stabilizing capacity provided by the foundation geometry.

6. GROUND ACCELERATION AND SAFETY FACTOR

The formula giving the maximum structural acceleration is:

$$M_{acting} = M_{resisting}$$

It is possible to compute the ground acceleration in both directions, in the case $T_C \leq T_1 \leq T_D$:

$$a_{SLV,x} = \frac{S_{e,x}}{S \cdot F_0} \cdot \frac{T_1}{T_C} \text{ and } a_{SLV,y} = \frac{S_{e,y}}{S \cdot F_0} \cdot \frac{T_1}{T_C}$$

Where: T_1 the first period of vibration of the structure; T_C and T_D the characteristic periods of the response spectrum; S a coefficient accounting for the subsoil category and topographic conditions; F_0 a factor corresponding to the maximum spectral amplification under horizontal acceleration.

The safety factor can be computed using:

$$f_{a,x}^* = \frac{a_g}{a_{SLV,x}} \text{ and } f_{a,y}^* = \frac{a_g}{a_{SLV,x}}$$

and with S =1.5g , T_C =0.393s , $T_{1,x}$ = 0.826s , $T_{1,y}$ = 0.875s , F_0 =2.517 , and a_g =0.121g

Table 9 - Safety factor per section for seismic force in x direction.

z [m]	$S_{e,x}$ [g]	$a_{SLV,x}$ [g]	$f_{a,x}^*$ [-]
0.00	0.516	0.287	2.37
7.16	0.654	0.363	3.01
8.68	0.640	0.356	2.94
10.18	0.754	0.420	3.47
12.91	0.758	0.422	3.49
16.65	1.180	0.657	5.43
19.73	1.474	0.820	6.78
22.02	2.132	1.187	9.80
24.22	2.854	1.589	13.12
25.56	3.190	2.177	17.98
26.82	7.209	4.013	33.15

Table 10 - Safety factor per section for seismic force in y direction.

z [m]	$S_{e,y}$ [g]	$a_{SLV,y}$ [g]	$f_{a,y}^*$ [-]
0.00^-	0.516	0.304	2.51
7.16	0.654	0.386	3.19
8.68	0.628	0.371	3.06
10.18	0.744	0.439	3.62
12.91	0.744	0.439	3.63
16.65	1.164	0.687	5.67
19.73	1.451	0.856	7.07
22.02	2.102	1.240	10.24
24.22	2.854	1.683	13.90
25.56	3.190	2.306	19.05
26.82	7.209	4.252	35.12

6.2. Analysis of the result

As previously discussed, the critical point in the tower's seismic stability is located in the first vertical section along the Y direction, where the resisting moment, in limit cases, is lower than the acting moment. In the present analysis, the behavior of the tower was studied assuming that this critical section near the base behaves as if walls are present continuously along the entire perimeter, similarly to the foundation section. This assumption is justified by the presence of a full masonry foundation slab and enables a more realistic estimate of the resisting moment in that zone.

Under this assumption, all calculated safety factors exceed unity, and the corresponding ground accelerations at the Life Safety Limit State (SLV) are significantly higher than those derived from the elastic horizontal spectra for both SLD and SLV. This means that the tower would theoretically withstand seismic actions associated with return periods well beyond 2475 years in both principal directions. Accordingly, both the seismic safety index and the acceleration factor were found to be greater than one, confirming the structural stability under the simplified global approach.

7. LOCAL ASSESSMENT – LV2 ANALYSIS OF THE CRITICAL SECTION

In addition to the global LV1 assessment, a local LV2 analysis was performed on the most critical portion of the bell tower, specifically the base section along the Y-axis, where the presence of a large arch significantly reduces the available resisting masonry area. The Level 2 analysis is particularly suited for the evaluation of seismic safety in cases involving local vulnerabilities or when strengthening interventions are considered on limited portions of the structure, known as macroelements. In this context, the arch and the adjacent wall piers at the tower base were identified as a single macroelement potentially prone to local collapse mechanisms.

The primary objective of an LV2 assessment is to determine the horizontal load multiplier (α_0) that activates a local collapse mechanism in the analyzed portion. This is especially relevant in historic masonry structures, where seismic failure often initiates from partial instability rather than from the global collapse of the entire structure.

The horizontal seismic forces were introduced as equivalent inertial actions, proportional to the vertical loads and to an unknown collapse multiplier α_0. By applying the Principle of Virtual Work to the assumed mechanism and computing the equilibrium between stabilizing and overturning moments, the minimum value of α_0 was obtained. This value represents the seismic capacity of the identified macroelement.

It is worth noting that this analysis assumes no tensile strength in the masonry, infinite compressive strength, and neglects sliding failure. The critical influence of wall openings and the configuration of the arch at the base were also taken into account, as they modify the lever arms and reduce the stabilizing effect of vertical loads.

Although this is a localized evaluation, the calculated collapse multiplier contributes to understanding the seismic vulnerability of the tower and complements the LV1 analysis by providing insight into a potential local failure mechanism. Furthermore, different confidence factors can be applied to the macroelement, depending on the knowledge level and the modeling approach. In this case, since a rigid-body mechanism was assumed, the confidence factor was applied directly to the resulting acceleration capacity.

7.1. Linear analysis and load multiplier α_0

A simple linear kinematic analysis was carried out by imposing a limit equilibrium condition at the onset of motion for the macroelement identified at the base of the tower. The collapse mechanism was modeled as the overturning of a rigid block representing the masonry pier adjacent to the large arch opening.

To determine the seismic capacity, the analysis focused on the condition in which the overturning moment induced by the seismic force equals the stabilizing moment provided by the vertical loads. This corresponds to the activation of the mechanism. The computational procedure follows the classical principle of virtual work, applied to the assumed mechanism.

The key output of the analysis is the load multiplier α, which scales the horizontal seismic action to the point where collapse is imminent. In this simplified model, the expression for α has a purely geometric solution, depending only on the location of the center of mass, the dimensions of the overturning block, and the lever arms of stabilizing and overturning forces. The following figure illustrates the idealized mechanism and the parameters used in the calculation.

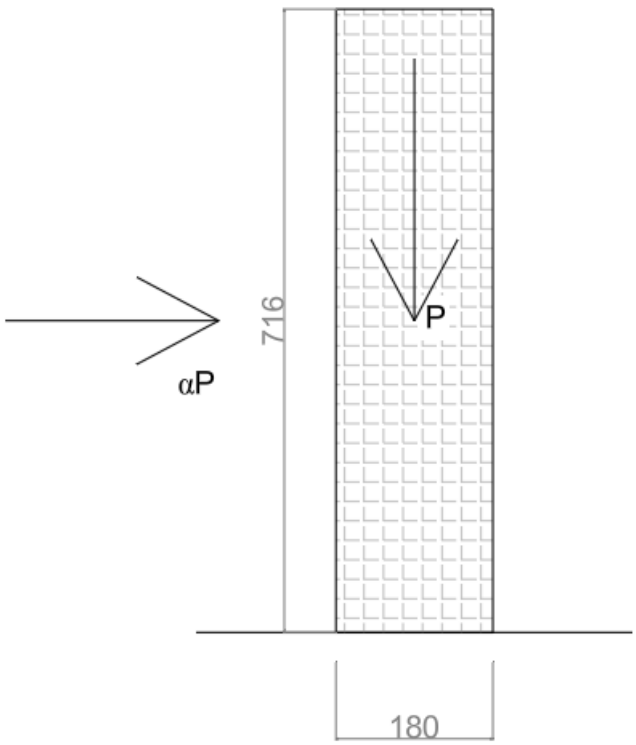

Figure 14- Vertical equilibrium scheme of the overturning mechanism. All dimensions are expressed in centimeters.

$$\alpha = \frac{B}{H} = 0.251397$$

Where B is the effective base of one of the two cross sections length of the overturning block along the Y-axis, equal to 3.60 m; H is the height of the critical section, equal to 7.16 m; P is half of the total vertical load applied on the section, which does not influence the value of the collapse multiplier.

In this simplified mechanism, the overturning is assumed to involve only one side pier of the arch, and thus the multiplier α represents the capacity of a single structural unit, not of the entire arch support system.

7.2. Resisting moment in the critical section.

To evaluate the resisting moment of the critical section, the equilibrium is considered under the limit condition in which only part of the section is active, due to the no-tension behavior of masonry. It is assumed that the load acts with an eccentricity equal to 7/18 of the base width of the pier. Under this assumption, the resisting moment provided by a single pier can be expressed as:

$$M_{rd}^{1} = N^{1} \cdot e = 20959kN \cdot m$$

$$e = \frac{7}{18} \cdot B = 2.917m$$

Where N is half of the total vertical load acting on the entire base section (as only one pier is considered at a time); e is the eccentricity of the load.

Since the system consists of two symmetrical piers flanking the arch, the total ultimate resisting moment of the section can be estimated as:

$$M_{rd}, 1 = 2 \cdot \alpha \cdot M_{rd}^{1} = 10538kN \cdot m$$

Where α is the previously computed collapse multiplier derived from the local kinematic analysis. This formulation allows a direct comparison between the resisting and acting moments at the base of the arch, validating the section's capacity against the seismic demand.

7.3. Analysis of the result

The ultimate resisting moment MRd,x in the first vertical section near the base of the tower, evaluated through the Level 2 analysis, was found to be 10538 kNm. This value is significantly lower than the corresponding acting moment Mx, which reaches 23024 kNm under the seismic scenario considered.

This result clearly indicates that the current configuration of the structure in this zone is insufficient to withstand the expected seismic demand along the Y-axis, and therefore strengthening interventions are necessary to ensure adequate seismic performance. The analysis performed using the LV2 methodology allowed for a more refined estimation of the actual resistance of the critical section, taking into account the influence of geometry, eccentric loading, and the local overturning mechanism of the masonry piers adjacent to the base arch.

Based on the updated resisting moment value, a revised interaction diagram was developed to visualize the comparison between the resisting and acting moments in the Y direction. This updated graph highlights the extent of the structural deficiency and supports the need for targeted retrofit measures.

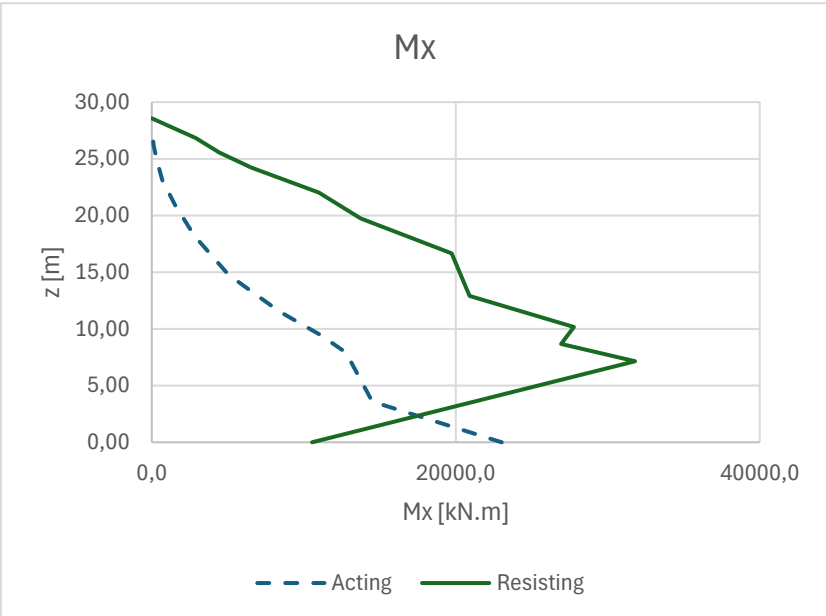

Figure 15 - Acting and resisting bending moments (Mx) along the height of the tower.

8. CONCLUSION

The analysis highlights a significant reduction in the resisting bending moment in the lower part of the tower, particularly due to the presence of the large arch at the base, which leads to a reduction in the effective resisting cross-section. Although wall openings exist at various levels of the structure, the most critical condition occurs in the first vertical section along the Y-axis, where the acting seismic moment significantly exceeds the available resistance.

While the global seismic assessment using the simplified LV1 method suggests that the tower performs adequately against the design seismic actions—yielding both an acceleration factor and seismic safety index greater than unity—this alone does not provide a complete picture. In fact, the refined local analysis performed through the LV2 approach has revealed that the critical base section is not capable of resisting the seismic demand without exceeding its ultimate moment capacity. This discrepancy between global capacity and local vulnerability underscores the importance of assessing partial collapse mechanisms in heritage structures.

Through LV2 analysis, a realistic estimation of the resisting moment in the critical section was obtained, and the updated moment comparison graph clearly indicates the need for strengthening interventions at the base. These could include reinforcement of the piers adjacent to the arch, improved wall connectivity, or other localized retrofitting strategies.

In conclusion, while the tower exhibits acceptable performance in terms of global stability, the local deficiency identified in the base section demands attention. Further investigations using LV3 methods could provide additional insight, but the results already support the recommendation for targeted structural interventions to enhance the tower's seismic safety.